

The Future Unemployed

The NEXT WAVE IN TECHNOlogy

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Figure 1: Team Logo

Manual Overview

Description

This manual provides a comprehensive guide for setting up and using the Panther Cloud Air (PCA) system. It covers the full installation process, including environment configuration and database setup, followed by a user guide for employees detailing how to search flights and manage bookings, and an administrator guide covering how to view route and aircraft profitability reports and update profit margins within the system.

Installation Guide

Hardware and Software Requirements

Hardware

- Model: Dell Pro 14 Premium
- Processor: Intel Core Ultra 7 268V, 2200 MHz, 8-core processor
- Memory: 32 GB of RAM

- Storage: 120 GB of available disk space
- Display: 14-inch monitor

Software

- Operating System: Microsoft Windows 11 Enterprise or equivalent
- Code Editor: Visual Studio Code or any preferred development environment
- Version Control: Git and Github
- Backend Server: Flask, deployed via Docker
- Database Management: MariaDB, deployed via Docker
- Containerization: Docker Desktop

Installation Overview

Before beginning installation, users should ensure they have Docker Desktop, Git, and a code editor such as Visual Studio Code installed on their device. The system is not limited to any specific operating system and can run on Windows, macOS, or Linux provided these prerequisites are met. Once prerequisites are in place, this section will walk through cloning the GitHub repository, environment configuration, Docker setup, Flask configuration, and database initialization to get the system up and running

Installation Instructions

Step 1: Install Prerequisites

Before setting up the Panther Cloud Air system, the following software must be installed on your device.

- Git — Download and install from https://git-scm.com/downloads. Git is required to clone the project repository to your local machine.
- Docker Desktop — Download and install from https://www.docker.com/products/docker-desktop. Docker Desktop is required to run the Flask backend and MariaDB database.
- Visual Studio Code — Download and install from https://code.visualstudio.com/download. Visual Studio Code is the recommended code editor for managing and running the project, though any preferred development environment may be used.

Step 2: Clone the Repository

The repository is the folder that contains all of the files needed to run the Panther Cloud Air system. Follow the steps below to download it to your device.

1. Open Visual Studio Code on your device.
2. Press Ctrl + `to open the terminal. A panel will appear at the bottom of the screen.
3. In the terminal panel, type the following command and press Enter: git clone git clone https://github.com/whodja/CSC-4710-Project.git
4. Once the download is complete, type the following command and press Enter to open the project folder: cd CSC-4710-Project

Step 3: Start Docker

Docker is the program that runs the backend of the Panther Cloud Air system. Follow the steps below to get it running.

1. Open Docker Desktop on your device and wait for it to fully load. You will know it is ready when the icon in your taskbar stops animating.
2. Return to Visual Studio Code and in the terminal panel at the bottom of the screen, type the following command and press Enter: docker-compose up --build
3. Docker will now set up everything the system needs to run. This may take a few minutes the first time. Do not close the terminal window while this is happening.
4. Once you see a message in the terminal that says the Flask server is running, the system is ready.

Step 4: Accessing the System

1. Open any web browser on your device, such as Google Chrome, Microsoft Edge, or Firefox.
2. In the address bar at the top of the browser, type the following address and Press Enter: http://localhost:5001
3. The Panther Cloud Air system will now load and is ready to use.

Employee User Guide

Administrator User Guide